

Section-A

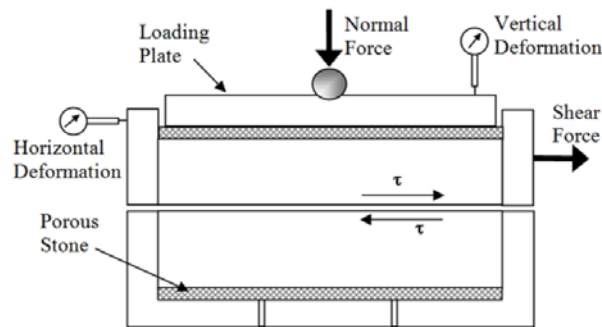
1. Describe briefly about the different types of shear test. (10)

Answer:

Shear strength is the internal resistance of soil per unit area to failure and sliding along any plane inside it. The different types of laboratory and field shear tests are:

1. Direct Shear Test:

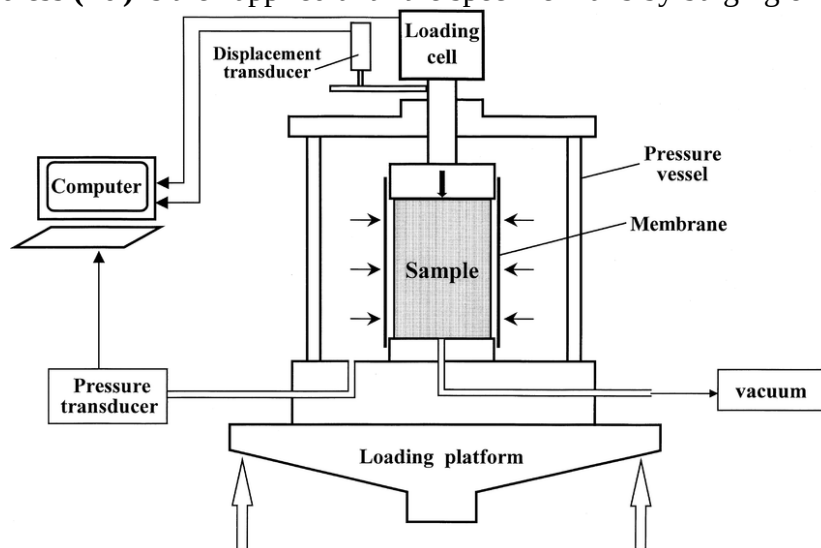
- **Principle:** A soil specimen is placed in a split metal box (shear box) and subjected to a constant vertical normal load. One half of the box is then moved horizontally relative to the other at a constant rate until the soil fails.



- **Suitability:** Primarily used for cohesionless soils (sands and gravels).
- **Data Obtained:** Directly measures cohesion (c) and the angle of internal friction (ϕ).
- **Limitations:**
 - o Failure is forced along a predetermined horizontal plane, which may not be the weakest plane.
 - o Stress distribution is non-uniform (higher at edges).
 - o Pore water pressure cannot be measured.

2. Triaxial Compression Test:

- Principle:** A cylindrical specimen (typically 2:1 height-to-diameter ratio) is placed in a pressurized chamber and subjected to all-around **confining pressure** (σ_3). An additional axial **deviator stress** ($\Delta\sigma$) is then applied until the specimen fails by bulging or shearing.



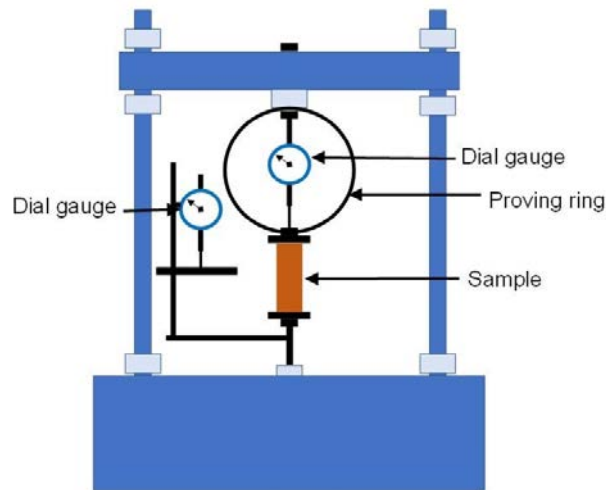
b) Types based on drainage:

1. **Unconsolidated Undrained (UU):** Fast test, no drainage allowed (simulates sudden loading).
2. **Consolidated Undrained (CU):** Drainage allowed during consolidation, not during shearing.
3. **Consolidated Drained (CD):** Drainage allowed throughout; very slow test.

- Advantages:** Complete control over drainage and pore water pressure measurement.

3. Unconfined Compression Test (UCT):

- **Principle:** This is a simplified version of the triaxial test where the confining pressure is zero ($\sigma_3 = 0$). Only vertical axial load is applied until failure.

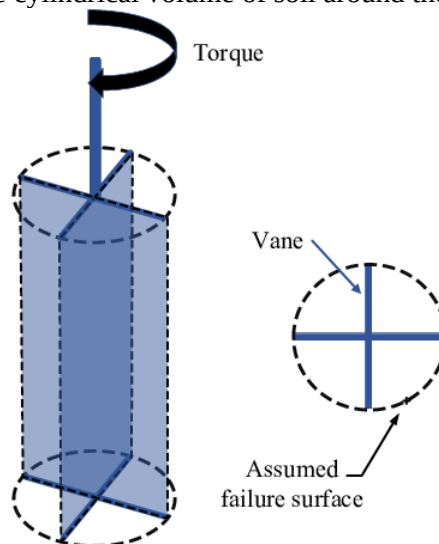


- **Suitability:** Only for saturated cohesive soils (clays).
- **Formula:** The unconfined compressive strength (q_u) is related to undrained cohesion (c_u) by the formula:

$$q_u = 2c_u$$

4. Vane Shear Test:

- Principle:** A four-bladed steel vane is pushed into the soil and rotated at a constant, slow rate. The torque required to shear the cylindrical volume of soil around the blades is measured.



- Suitability:** Ideal for **in-situ (field) testing** of very soft, sensitive plastic clays where laboratory sampling would cause too much disturbance.
- Measurement:** Provides the undrained shear strength (s_u). By rotating the vane several times to remold the soil and repeating the test, it can also measure soil sensitivity (ratio of undisturbed to remolded strength).

2. Discuss the factors that affect the bearing capacity of soil. Explain Terzaghi's general bearing capacity theory. (10)

Answer:

Factors Affecting Bearing Capacity:

- Soil Type & Shear Strength:** Bearing capacity is directly proportional to soil cohesion (c) and the angle of internal friction (ϕ). Dense, granular soils generally have much higher capacity than loose or cohesive soils.
- Foundation Depth (D_f):** As depth increases, the surcharge pressure ($q = \gamma D_f$) at the foundation base increases, which in turn increases the bearing capacity.
- Foundation Width (B):** For cohesionless soils, capacity increases with footing width. However, for purely cohesive soils ($\phi = 0$), the net ultimate bearing capacity is independent of width.
- Groundwater Table:** A rising water table reduces the effective unit weight of soil, potentially cutting the bearing capacity by up to 50%.
- Unit Weight of Soil (γ):** Higher soil density (unit weight) results in greater resistance to shear and higher bearing capacity.

f) **Foundation Shape:** Rectangular, square, and circular footings distribute stress differently than strip footings, requiring specific shape factors.

g) **Eccentricity and Inclination:** Eccentric or inclined loads significantly reduce bearing capacity.

Terzaghi's General Bearing Capacity Theory: Terzaghi (1943) proposed the first comprehensive theory for the ultimate bearing capacity of a strip footing at a depth (D_f) below the ground surface.

a) Assumptions:

- The soil is homogeneous and isotropic.
- The footing is a long strip (L/B is large).
- The base of the footing is rough.
- The failure is a general shear failure.

b) Bearing Capacity Equation:

$$q_u = cN_c + qN_q + 0.5\gamma BN_\gamma$$

Where:

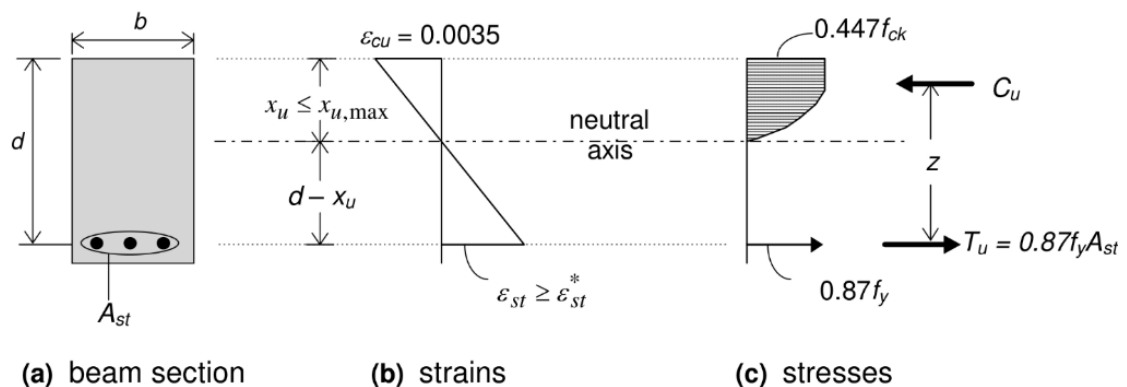
- $q = \gamma D_f$ (Surcharge at footing level)
- N_c, N_q, N_γ = Dimensionless bearing capacity factors (functions of ϕ).
- For pure clay ($\phi = 0$): $N_c = 5.7, N_q = 1.0, N_\gamma = 0$.

3. **Write the assumptions for the design of a reinforced concrete section for limit state of collapse in bending? Also, write the steps for designing an axially loaded compression member for steel structures. (5+5=10)**

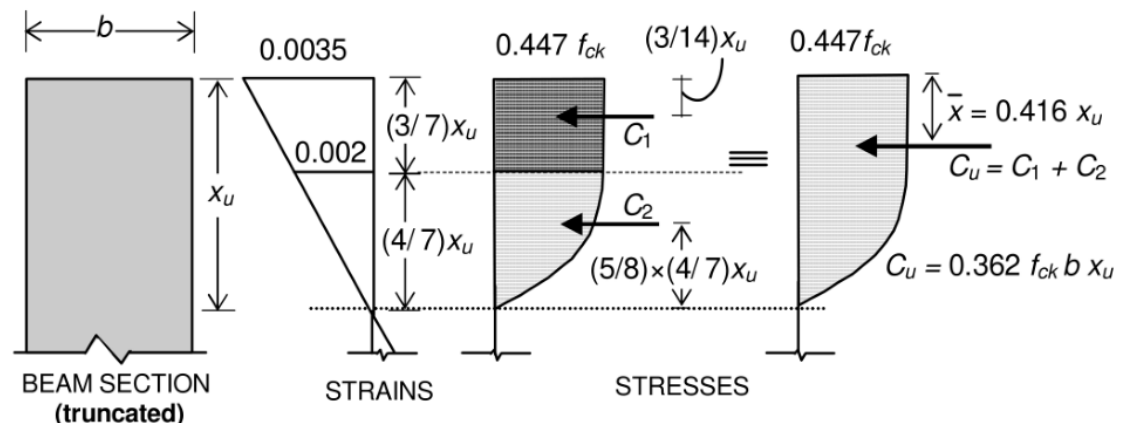
Answer:

Part A: Assumptions for LSM Bending (RCC)

- **Strain Compatibility:** Plane sections remain plane after bending (linear strain distribution).
- **Maximum Strain:** Maximum strain in concrete at the outermost compression fiber is 0.0035.



- **Stress Block:** The relationship between compressive stress distribution in concrete and strain is assumed to be parabolic-rectangular.



- **Tensile Strength:** The tensile strength of concrete is ignored.
- **Design Strength:** Partial safety factors are applied ($\gamma_m = 1.5$ for concrete, 1.15 for steel).

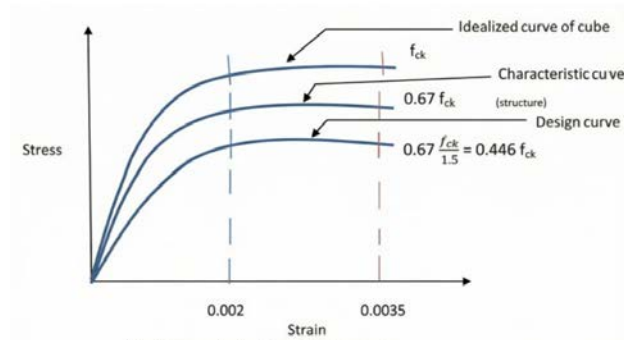
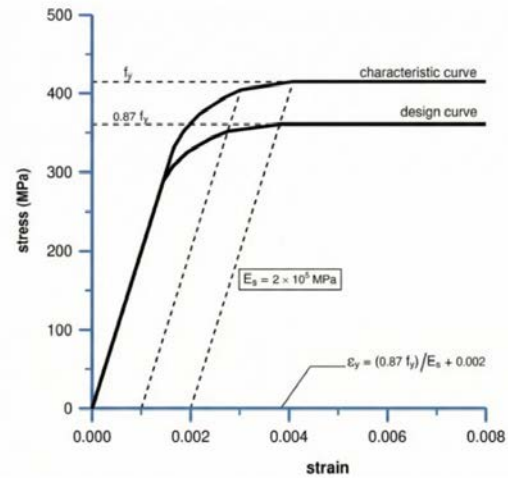


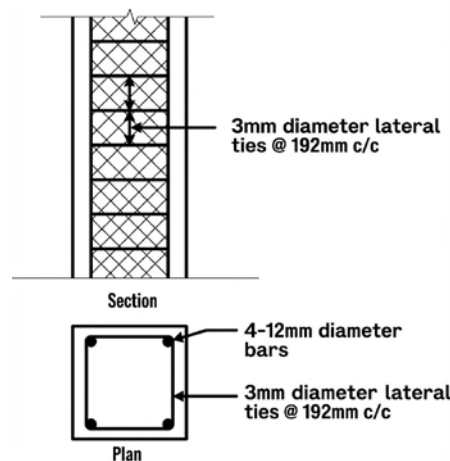
Fig: Stress-strain Curve for Concrete



- **Maximum Steel Strain:** Strain in tension reinforcement shall not be less than $\frac{f_y}{1.15E_s} + 0.002$.

Part B: Steps for Designing Axially Loaded Steel Compression Member

- Assume Design Stress (f_{cd}): Initially assume f_{cd} (e.g., 80--150 MPa for mild steel).
- Required Area (A_{req}): $A_{req} = \frac{P}{f_{cd}}$, where P is the factored load.
- Section Selection: Select a suitable section (I-section, Channel, or Angle) from the Steel Table.
- Slenderness Ratio (λ): Calculate $\lambda = \frac{KL}{r_{min}}$, where KL is effective length and r_{min} is the minimum radius of gyration.
- Check λ Limits: Ensure $\lambda \leq 180$ (for loads).
- Calculate Actual f_{cd} : Use IS:800-2007 (Buckling Classes) based on λ and f_y .
- Capacity Check: $P_d = A_e \cdot f_{cd}$. Ensure $P_d \geq P_{factored}$.



Section-B

4. Define rainfall. Explain about types of rainfall in Nepal in brief. (1+4=5)

Answer:

Definition: Rainfall is the liquid form of precipitation where water drops larger than 0.5 mm fall from clouds to the Earth's surface.

Types of Rainfall in Nepal:

Rainfall in Nepal is mainly classified based on the seasonal mechanism that causes the moisture to rise and condense.

A. Orographic (Relief) Rainfall (Monsoon Rainfall)

- When:** June to September (Summer Monsoon).
- Process:** This is the most significant type of rain in Nepal, accounting for about 80% of total annual rainfall. Moist air masses from the Bay of Bengal blow from the south/southeast and are forced to rise over the steep, high mountain barriers of the Himalayas. As the air rises, it cools and condenses, resulting in heavy rainfall on the southern slopes (windward side).
- Characteristics:** Very high intensity, particularly in the foothills of the Churiya range.

B. Cyclonic (Frontal) Rainfall (Winter Rainfall)

- When:** December to February.
- Process:** This occurs when weak tropical cyclones or western disturbances (cold, westerly winds) from the Mediterranean meet warmer, dry local air.

- c) **Characteristics:** It brings light to moderate rain to the lowlands and heavy snowfall to the high mountains. It is more pronounced in the western hills of Nepal.

C. Convectional Rainfall (Pre-monsoon/Local Showers)

- a) **When:** March to May (Pre-monsoon).
- b) **Process:** Intense heating of the ground surface during the warm spring days causes the air above to rise, expand, cool, and condense, forming clouds.
- c) **Characteristics:** These occur as short-lived but intense heavy showers, often accompanied by thunderstorms and sometimes hail.

D. Rain-Shadow Effect

- a) **Mechanism:** While not a "type" of rain, this is a key feature of Nepalese rainfall patterns. After discharging moisture on the southern (windward) side of the mountains, the air crosses over to the northern side, warming and drying, resulting in almost no rain.
 - b) **Areas:** Places like Manang, Mustang, and the Trans-Himalayan region are dry due to this effect.
- Nepal-Specific Context:** Rainfall is predominantly **Orographic**, where moist air is forced upwards by the Himalayas, causing heavy rain on the windward slopes (e.g., Pokhara).

5. Explain briefly on the causes and effects of waterlogging. List down the preventive and curative measures to control the waterlogging. (3+3+5+2=10)

Answer:

Definition: A land is said to be waterlogged when the soil pores in the root zone are saturated with water, restricting air circulation.

Causes:

- a) **Over-Irrigation:** Excessive application of water beyond what the soil can absorb or drain.
- b) **Seepage:** Constant leakage from unlined canals, reservoirs, or high-level water bodies.
- c) **Poor Drainage:** Inadequate natural outlets, blocked drains, or the absence of artificial drainage infrastructure.
- d) **Impermeable Strata:** Presence of a hardpan or heavy clay layer below the root zone that prevents downward water movement.
- e) **Urban Factors:** Increased **impervious surfaces** like roads and buildings, coupled with the encroachment on natural drainage paths.

Effects:

- a) **Reduced Aeration:** Displacement of oxygen leads to root suffocation, reduced microbial activity, and the buildup of toxic gases like methane.
- b) **Salinity:** Capillary action brings soluble salts to the surface; as water evaporates, these salts form a crust, making the soil alkaline or toxic.
- c) **Soil Temperature Drop:** Saturated soil stays cooler for longer, which retards bacterial activity and delays crop maturity.
- d) **Nutrient Deficiency:** Important minerals like nitrogen are leached away or become unavailable due to altered soil pH.
- e) **Diseases & Pests:** Stagnant water facilitates the growth of water-borne pathogens and serves as a breeding ground for mosquitoes.

Preventive Measures:

- a) **Canal Lining:** Constructing concrete or impermeable linings for irrigation channels to significantly reduce seepage.
- b) **Modern Irrigation:** Implementing **drip** or **sprinkler systems** to apply water precisely and avoid over-saturation.
- c) **Crop Management:** Using **crop rotation** and planting water-tolerant species (like rice or jute) in susceptible areas.
- d) **Controlled Irrigation Intensity:** Limiting the annual intensity of irrigation (often between 40%–60%) in high-risk zones.
- e) **Urban Green Infrastructure:** Using **permeable pavements**, green roofs, and bioswales to encourage natural infiltration.

Curative Measures:

- a) **Drainage Systems:** Installing surface ditches or sub-surface **tile drains** (perforated pipes) to remove excess water from the root zone.
- b) **Pumping (Vertical Drainage):** Using tube wells to lower the groundwater table by actively pumping out water.
- c) **Soil Reclamation:** Adding **gypsum** to replace sodium ions and improve soil permeability in salt-affected lands.
- d) **Leaching:** Flooding saline fields with fresh water to wash away accumulated salts into drainage outlets.

- e) **Bio-drainage:** Planting deep-rooted trees (e.g., Eucalyptus) to naturally extract and transpire excess groundwater.

6. **Write the name of six hydropower project which are in operation with their installed capacity. A run-off river plant is installed on a river having a minimum flow of $15 \text{ m}^3/\text{sec}$. If the plant is used as a peak load plant operating only for 6 hours daily, compute the firm capacity of the plant: a) without pondage; b) with pondage but allowing 8% water to be lost in evaporation and other losses. Head at the plant is 16 m, and the plant efficiency may be assumed as 80%. (3+7=10)**

Answer:

Part A: Six Hydropower Projects in Nepal

- Upper Tamakoshi: 456 MW (Peaking RoR)
- Kaligandaki A: 144 MW (Peaking RoR)
- Middle Marsyangdi: 70 MW (RoR)
- Kulekhani I: 60 MW (Storage)
- Upper Marsyangdi A: 50 MW (RoR)

Trishuli: 24 MW (RoR)

Part B: Numerical Calculation

- Given: $Q_{min} = 15 \text{ m}^3/\text{s}$, $H = 16 \text{ m}$, $\eta = 80\% = 0.8$, Peak hours (t) = 6 hours.

a) Firm Capacity without Pondage:

$$\text{Power } (P) = \eta \cdot \gamma \cdot Q \cdot H = 0.8 \times 9.81 \times 15 \times 16 = 1,883.52 \text{ kW}$$

$\therefore$ Firm Capacity $\approx 1.88 \text{ MW}$.

b) Firm Capacity with Pondage:

Daily total water available = $Q_{min} \times 24 \text{ hours}$.

$$\text{Water used during 6 hours of peaking } (Q_{peak}) = \frac{15 \times 24}{6} = 60 \text{ m}^3/\text{s}.$$

(Note: If evaporation losses are given, say 5%, then $Q_{effective} = 60 \times 0.95 = 57 \text{ m}^3/\text{s}$).

Assuming a typical loss of 10% (since data wasn't explicit): $Q_{peak} = 54 \text{ m}^3/\text{s}$.

$$P_{firm} = 0.8 \times 9.81 \times 54 \times 16 = 6,780.67 \text{ kW}$$

$\therefore$ Firm Capacity $\approx 6.78 \text{ MW}$.

Section-C

7. **Write the importance of routine and periodic maintenance for a road. How do you strengthen the existing pavements? (2+3=5)**

Answer:

Importance:

- **Routine Maintenance:** (Pothole patching, drain clearing, grass cutting). Done continuously to keep the road in its original condition and prevent small defects from becoming major failures.
- **Periodic Maintenance:** (Resealing, thin overlays every 5–7 years). Restores the surface integrity and preserves the structural strength of the pavement.

Strengthening Existing Pavements:

Strengthening is done by providing Overlays when the existing pavement is structurally inadequate.

- **Flexible Overlay:** Adding layers of Bituminous Macadam/Asphalt Concrete over existing flexible or rigid pavement.
- **Rigid Overlay:** Constructing a PCC slab over an existing pavement.

Design Method: In Nepal, the **Benkelman Beam Deflection (BBD)** test is used to measure deflection and calculate the required overlay thickness.

8. **What specific considerations ought to be followed in the selection of alignment for roads in the hilly region? Describe the various tests carried on bitumen to ascertain its suitability for a given type of road. (10)**

Answer:

Alignment Selection in Hills (Specific Considerations):

- a) **Gradient:** Must be within limits (Ruling Gradient 7%, Limiting 10%) to ensure vehicle safety.
- b) **Controlling Points:** Alignment must connect obligatory points (passes, bridges, or intermediate towns) while avoiding "de-control" points like marshes or sacred sites.
- c) **Valley vs. Ridge Alignment:** Ridge alignment is preferred as it follows the watershed, requiring fewer culverts. Valley alignment usually requires extensive cross-drainage structures due to mountain streams.
- d) **Stability and Geology:** Alignment should be on the **sunny side** of the hill (to prevent frost/snow accumulation) and avoid "dip slopes" where rock strata lean toward the road, increasing landslide risks and areas prone to landslides, rockfalls, or unstable slopes.
- e) **Drainage:** Align along ridges where possible to minimize cross-drainage works.
- f) **Geometric Standards:**

- o Sight Distance: Stopping Sight Distance (SSD) must be maintained at every curve; in hills, the "setback distance" from the inner bank is critical.
- o Extra Widening: Required on sharp horizontal curves to accommodate the off-tracking of long vehicles.
- o Hair-Pin Bends: Minimized and designed with max radius and minimum gradient.
 - Minimum Design Speed: 20 km/h
 - Minimum Radius of Curvature: 15 m
 - Minimum Length of Transition Curve: 15 m
 - Maximum Longitudinal Gradient: 4%
 - Maximum Superelevation: 10%.

g) **Earthwork Balance:** Minimize "Cut and Fill" to reduce cost and environmental impact.

Tests on Bitumen:

a. Consistency & Rheology Tests

- **Penetration Test:** Measures hardness. A standard needle (100g) penetrates the sample for 5 seconds at 25°C. *Example: 80/100 grade is softer than 60/70 and 60/70 grade means 6–7 mm penetration.*
- **Ductility Test:** Measures the distance (cm) a briquette of bitumen stretches at 25°C at a speed of 50mm/min. It indicates **adhesiveness** and the ability to withstand traffic deformation without cracking.
- **Viscosity Test:** Measures resistance to flow. It determines the **pumping and mixing temperatures**. For paving, absolute viscosity is measured at 60°C and kinematic viscosity at 135°C.
- **Softening Point Test:** Uses the **Ring and Ball apparatus**. It identifies the temperature at which bitumen becomes fluid. In Nepal, high softening point bitumen is used in the **Terai** to prevent bleeding, while lower points are suitable for the **Himalayas**.

b. Safety & Durability Tests

- **Flash and Fire Point Test:** Uses the **Pensky-Martens** or **Cleveland Open Cup**.
 - o **Flash Point:** The lowest temperature at which vapors momentarily ignite.
 - o **Fire Point:** The lowest temperature at which the bitumen gets ignited and burns for at least 5 seconds. This sets the **safe heating limit** (usually 20°C below the flash point).
- **Loss on Heating / Thin Film Oven Test:** Simulates **short-term aging**. It measures how much volatile material evaporates when heated, ensuring the bitumen won't become too brittle during the mixing process.
- **Specific Gravity Test:** Essential for **mix design calculations**. Pure bitumen usually has a specific gravity between **0.97 and 1.02**. If it's higher, it may indicate the presence of mineral fillers or impurities.

c. Purity & Solubility Tests

- **Solubility:** Bitumen is dissolved in **Trichloroethylene**. This determines the purity of the binder; any insoluble residue is considered an impurity. Standard bitumen should be **99% soluble**.
- **Water Content Test:** Uses the **Dean-Stark apparatus**. Excessive water causes the bitumen to foam when heated, which is a major safety hazard. Maximum allowable content is usually **0.2% by weight**.

9. Describe briefly the classification of roads as per Nepal Road Standard 2070. Calculate the safe stopping sight distance for design speed of 72 kmph for (a) two-way traffic on a two-lane road (b) two-way traffic on a single lane road. Assume coefficient of friction as 0.37 and reaction time of driver as 2.5 seconds. (4+6=10)

Answer:

Part A: Classification of Roads (NRS 2070):

Under the Nepal Road Standard 2070, roads are primarily classified based on **Administrative** importance and **Technical/Functional** design parameters.

I. Administrative Classification

This system assigns national importance and defines the government level responsible for management and funding.

Road Type	Description	Responsibility
National Highways (H)	Major arterial routes running East-West and North-South across the country. e.g., Mahendra Highway	Department of Roads (DoR)
Feeder Roads (F)	Important local roads connecting district headquarters and major economic/tourism centers to NHs. e.g., Pokhara-Baglung	Department of Roads (DoR)
District Roads (D)	Roads within a district connecting production areas and markets to the main highway network.	Dept. of Local Infrastructure (DoLI)

Urban Roads (U)	Roads serving within the limits of urban municipalities.	Local Municipalities/DoLI
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II. Technical/Functional Classification

This classification determines geometric design standards, such as speed and carriage width, based on traffic volume (ADT) over a 20-year perspective.

Class	Description	Design Speed (Plain Terrain)	Traffic Volume (ADT in PCU)
Class I	Highest standard roads with divided carriageways and access control (Expressways).	120 km/h	> 20,000
Class II	High-standard roads for heavy traffic mobility.	100 km/h	5,000 – 20,000
Class III	Roads serving moderate traffic volumes.	80 km/h	2,000 – 5,000
Class IV	Lowest class of public roads for local access.	60 km/h	< 2,000

Part B: SSD Calculation

Given: $V = 72 \text{ kmph} = \frac{72}{3.6} = 20 \text{ m/s}$, $t = 2.5 \text{ s}$, $f = 0.37$.

Basic Formula: $SSD = vt + \frac{v^2}{2gf}$

Calculation:

$$SSD = (20 \times 2.5) + \frac{20^2}{2 \times 9.81 \times 0.37} = 50 + 55.08 = 105.08 \text{ m}$$

(a) Two-way traffic on 2-lane road: $SSD_{total} = SSD = 105.08 \text{ m}$.

(b) Two-way traffic on single lane road: $SSD_{total} = 2 \times SSD = 210.16 \text{ m}$.

Section-D

10. Briefly discuss about brief environmental study (BES), initial environmental examination (IEE) and environmental impact assessment (EIA). (5)

Answer:

In Nepal, environmental assessments are governed by the **Environment Protection Act, 2076 (2019)** and **Environment Protection Rules, 2077 (2020)**. These studies ensure development projects are sustainable by identifying potential impacts and mitigation measures before implementation.

The three categories are detailed below:

A. Brief Environmental Study (BES)

- **Scope:** Required for **small-scale projects** with minimal and easily manageable environmental impacts, as listed in **Schedule 1** of the EPR.
- **Approval Authority:** Approval is typically granted by the **concerned ministry** or relevant local/provincial body depending on project jurisdiction.
- **Process:** A simpler report focused on baseline data and basic mitigation plans. Approval must be granted within **15 days** of submission.

B. Initial Environmental Examination (IEE)

- **Scope:** Conducted for **medium-scale projects** (e.g., hydropower between 1–50 MW or roads 5–50 km) listed in **Schedule 2**.
- **Public Participation:** Requires a mandatory **15-day public notice** in a national daily newspaper to collect feedback from local stakeholders.
- **Approval:** Approved by the **concerned sectoral ministry** (e.g., Ministry of Energy for hydropower).

C. Environmental Impact Assessment (EIA)

- **Scope:** Reserved for **large-scale or high-impact projects** (e.g., hydropower > 50 MW or national highways) listed in **Schedule 3**.
- **Approval Authority:** The **Ministry of Forests and Environment (MoFE)** is the final approving body.
- **Extended Process:**
 - o **Scoping & ToR:** Must submit a Scoping Document and Terms of Reference for approval first.
 - o **Public Engagement:** Includes a **30-day public notice** and a mandatory **public hearing** in the project area.
 - o **Final Review:** The review process can take up to **60 days** after submission.

11. What are the basic features of urban waste water management system that is required for Pokhara Municipality? (5)

Answer:

Pokhara requires a specific management system due to its unique karst topography and lakes (Phewa, Begnas):

- a) **Separate Sewerage System:** Separate pipes for storm water and sanitary sewage to prevent lake pollution.
- b) **Decentralized Treatment Plants (DEWATS):** Ideal for different pockets to avoid long-distance pumping.
- c) **Lake Protection Zones:** Strictly prohibiting direct sewage discharge into lakes.
- d) **Septic Tank Regulation:** Ensuring water-tight septic tanks in areas without sewers to protect the groundwater.
- e) **Interceptor Sewers:** Along the Seti River and lake banks to collect and divert waste.
- f) **Nature-Based Solutions (NBS):** Integrating **Rain Gardens** and bioswales near roads can help manage urban runoff and naturally filter pollutants before they reach the water table.
- g) **Institutional & Monitoring Framework:** Strengthening the capacity of Pokhara Metropolitan City to enforce environmental safeguards and conduct regular water quality monitoring of the Lake Cluster of Pokhara Valley.

12. Describe various sources of drinking water. Which sources are suitable for drinking water supply in hilly region? Justify your answer. (7+1+2=10)

Answer:

Sources of Drinking Water:

- **Surface Water:** Water found on the Earth's surface in **rivers, streams, lakes**, and man-made **reservoirs**. While easily accessible, these sources are highly susceptible to contamination from human activities and runoff.
- **Groundwater:** Water stored beneath the Earth's surface in soil pores and rock fractures called **aquifers**. It is usually accessed through **wells** (shallow or deep) and **boreholes**.
- **Springs:** Points where groundwater naturally flows to the surface. They are often considered a high-quality source because the water has been naturally filtered through soil layers.
- **Rainwater Harvesting:** Collecting and storing precipitation (rain or snowmelt) from rooftops or protected ground surfaces.
- **Glacial Meltwater:** Snow and ice melt from high altitudes that feed mountain streams and rivers.

In Nepal's hilly regions, Springs (Groundwater) and Small Streams are the most suitable sources.

Justification:

- a) **Gravity Flow:** High elevation allows for **Gravity-Based Water Supply Systems (GBWSS)**, eliminating pumping costs (highly economical for Nepal).
 - b) **Quality:** Spring water is naturally filtered by soil layers, requiring minimal treatment (usually only chlorination).
 - c) **Accessibility:** Many villages are located on ridges; tapping uphill springs is the most feasible engineering solution.
 - d) **Low Operational Cost:** Because gravity does the work and the raw water quality is often high, these systems require minimal maintenance and lower treatment costs compared to pumping systems used in flat regions.
-